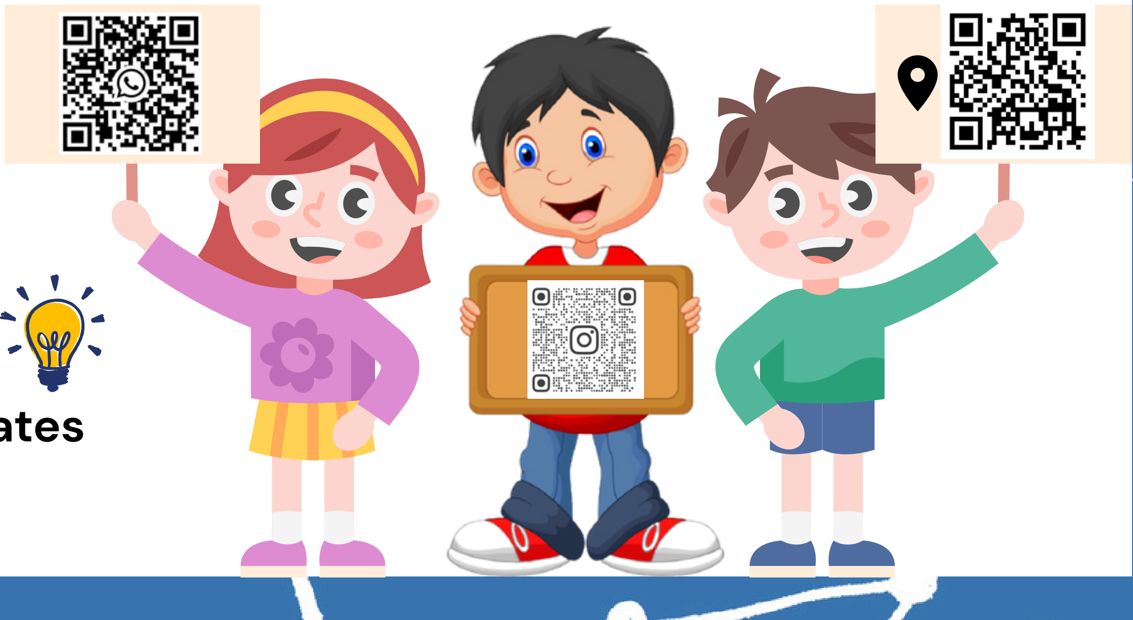
SATGURU STUDY CENTRE
Chennai - 81

TN CHENNAI DISTRICT STATEBOARD QUESTION PAPER 2024 - 2025



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Class : 12

Number

UNIT TEST - 2, AUGUST - 2024

Time Allowed : 1.30 Hours]

PHYSICS

[Max. Marks : 35

PART – I

I. Choose the correct Answer. 5x1=5

1. The flux linked with a coil at any instant t is given by $\phi_B = 10t^2 - 50t + 250$. The induced emf at t = 3s is

(a) -190 V (b) -10 V (c) 10 V (d) 190 V

2. In a transformer, the number of turns in the primary and the secondary are 410 and 1230 respectively. If the current in primary is 6A, then that in the secondary coil is

(a) 2A (b) 18 A (c) 12A (d) 1 A

3. Which of the following Electromagnetic Radiations is used for viewing objects through Fog.

(a) Microwave (b) Gamma Rays (c) X - rays (d) Infrared

4. Fraunhofer Lines are an example of ------------ spectrum.

(a) Line Emission (b) Line Absorbtion (c) Band Emission (d) Band Absorbtion

5. Q - Factor is equal to

(a) $\frac{\omega_r L}{R}$ (b) $\frac{1}{R}\sqrt{\frac{L}{C}}$ (c) $\frac{X_L}{R}$ (d) All the above

PART - II

II. Answer any Four questions and Question number 11 is compulsory. 4x2=8

6. Mention the ways of producing Induced emf.

7. How will you define RMS value of an Alternating Current?

8. A Straight metal wire crosses a Magnetic Field of Flux 4 mWb in a time 0.4 S. Find the magnitude of the emf induced in the wire.

9. What is Displacement Current?

10. Give two uses of UV Radiation.

11. The relative magnetic permeability of the medium is 2.5 and the relative electrical permittivity the medium is 2.25. Compute the refractive index of the medium.

PART- III

III. Answer any Four questions and question No. 17 is compulsory. **4x3=**

12. How will you induce an emf by changing the area enclosed by the Coil.
13. Obtain an expression for energy stored in an Inductor.
14. A 400 mH coil of negligible resistance is connected to an AC circuit in which an effective curren of 6 mA is flowing. Find out the voltage across the coil if the frequency is 1000 Hz.
15. Write down the Properties of Electromagnetic Waves. (any six)
16. Write a short notes on a) Micro wave b) X - ray
17. A magnetron in a microwave oven emits Electromagnetic waves (em waves) with frequency f = 2450MHz. What magnetic field strength is required for electrons to move in circular paths with this frequency?

PART - IV

IV. Answer all the questions. **2x5=10**

18. a) Derive an expression for Phase angle between the applied Voltage and Current in a series RLC Circuit.

(OR)

b) Explain the Construction and Working of a Transformer.

19. a) Write down Maxwell equations in Integral Form.

(OR)

b) Explain the types of Emission Spectrum.